AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM256
Cyber Security

Midterm Examination (MTA) - Makeup
Spring Semester 2024/2025

Date: 24/April/2025

Number of Exam Pages: 3 (including this cover sheet)	Time Allowed: 2 Hours

Instructions:

- Total Marks: **30**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **not** allowed.

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PART 1: Multiple Choice Questions [10 Marks]

Question 1: Choose the correct answer for the following statements:

1. Biometric authentication relies on an individual's unique aspect, such as a fingerprint or face. This method is best described as:
   a) Something you know
   b) Something you have
   c) Something you are
   d) Something you do

2. An advertising email sent to a company's customers displays every recipient's email address in the "From" line. This is a breach of:
   a) Availability
   b) Integrity
   c) Confidentiality
   d) Authentication

3. Out-of-band authentication (OOBA) involves:
   a) Using a single channel to exchange confidential factors
   b) Using multiple channels for authentication
   c) Encrypting passwords before transmission
   d) Disabling second-factor authentication

4. What is a critical asset in cybersecurity?
   a) Any hardware component in an organization
   b) An asset without which an individual or organization cannot function
   c) A financial investment in cybersecurity tools
   d) A non-essential system component

5. An unhappy employee copies confidential medical records to sell them to a newspaper. This is an example of:
   a) External cybercrime
   b) Insider human threat
   c) Accidental exposure
   d) Social engineering

6. Two-factor authentication (2FA) enhances password security against which types of attacks?
   a) Brute-force attacks and dictionary attacks
   b) Man-in-the-middle attacks
   c) Zero-day vulnerabilities
   d) Ransomware attacks

7. What is the purpose of server hardening?
   a) Improving the server's response time
   b) Securing server configuration to reduce vulnerabilities
   c) Increasing the number of users that can access the server
   d) Enhancing physical server durability

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8. Which security control category aims to restore systems after an incident?
   a) Preventive
   b) Detective
   c) Corrective
   d) Recovery

9. Which action is an example of an unintentional insider threat?
   a) Deliberately leaking sensitive data
   b) Forgetting to encrypt a USB drive
   c) Installing ransomware for financial gain
   d) Selling intellectual property to a competitor

10. To achieve complete system security, each component of a system must be:
    a) Secured individually
    b) Secured only at the network level
    c) Secured only at the endpoint level
    d) Left unsecured if it is not critical

PART 2: Open-Ended Questions [11 Marks]

Question 2: Pentesting is divided into three categories based on the testers' knowledge of the system they are examining. Identify these pen-testing categories with a brief description for each one. **[3 Marks]**

Question 3: Discuss the limitations of symmetric encryption systems. **[3 Marks]**

Question 4: Explain the concept of server hardening and why it is essential. **[3 Marks]**

Question 5: Explain the "The Simplified NCSC Kill Chain" model. **[2 Marks]**

PART 3: Problems [9 Marks]

Question 6: A system administrator discovers unusual activity on a Linux server. They suspect a potential security breach. They know that "Server hardening" is a good practice. Based on the information about **Server hardening** and **Keeping Linux up to date**, develop a plan for the system administrator to investigate and mitigate the potential breach. What additional security measures might you advise them to implement to prevent future incidents? **[4 Marks]**

Question 7: A fintech startup's payment API has no rate limit, allowing brute-force attacks. A dissatisfied DevOps engineer leaks API keys on GitHub, enabling attackers to drain $2M from user accounts. The company uses monolithic logging with no real-time alerts. **[5 Marks]**

1. Identify three systemic failures using the CIA triad.
2. Design a zero-trust architecture addressing the API and insider risks.

End of questions

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